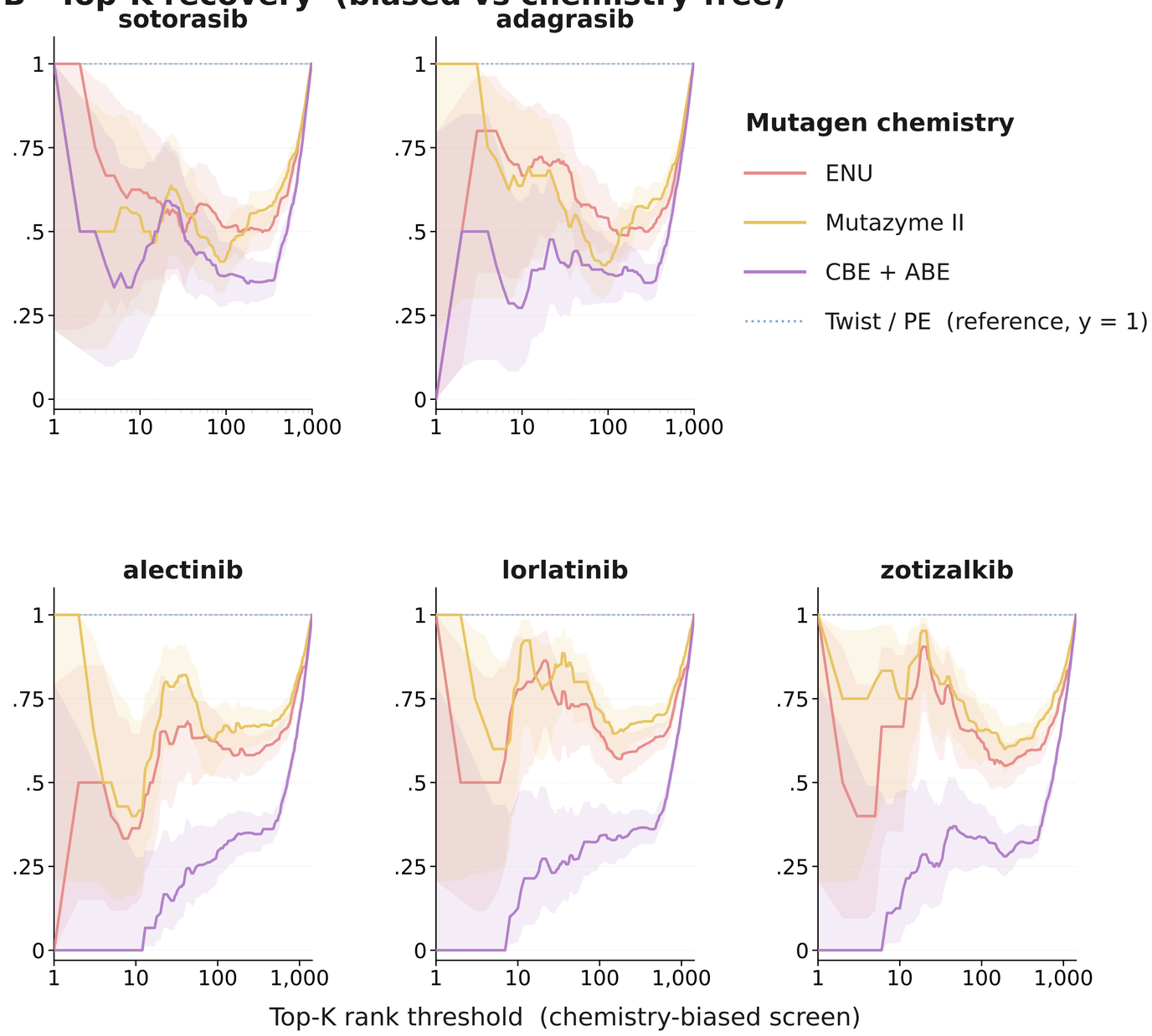
A Resistance panel × chemistry reachability

	ENU	Mutazyme II	CBE + ABE	Twist / PE
C12F	9.0%	18.0%	0.0%	16.7%
R68L	✗	✗	✗	✗
Y96C	36.6%	18.0%	50.0%	16.7%
V8L	9.0%	18.0%	0.0%	16.7%
V8M	✗	✗	✗	✗
Y96D	5.0%	3.6%	0.0%	16.7%
Y96H	36.6%	18.0%	50.0%	16.7%
H95D	0.2%	5.0%	0.0%	16.7%
H95Q	31.4%	10.5%	0.0%	16.7%
H95R	36.6%	18.0%	50.0%	16.7%
R68S	9.0%	18.0%	0.0%	16.7%
R68M	9.0%	18.0%	0.0%	16.7%
V1180L	0.2%	5.0%	0.0%	16.7%
L1196M	9.0%	18.0%	0.0%	16.7%
I1171N	31.4%	10.5%	0.0%	16.7%
I1171T	36.6%	18.0%	50.0%	16.7%
G1202R	17.9%	47.0%	50.0%	16.7%
G1269A	0.2%	5.0%	0.0%	16.7%
F1174C	5.0%	3.6%	0.0%	16.7%
F1174L	36.6%	18.0%	50.0%	16.7%
L1196Q	31.4%	10.5%	0.0%	16.7%
L1198F	17.9%	47.0%	50.0%	16.7%
C1156Y	17.9%	47.0%	50.0%	16.7%
L1152R	5.0%	3.6%	0.0%	16.7%

Chemistry-accessible (class supply ≥ 10 %)
Chemistry-under-counted (class supply < 10 %)
SNV-unreachable (2+ nt from WT codon)

B Top-K recovery (biased vs chemistry-free)



C Per-residue reachability of the top-ranked substitution

